Context-Aware Background Music Descriptor Generation from Movie Scripts

Harshal Karangale[1] **Suyash Pande**[1] **Chaitanya Kadupukutla**[1] **Shivam Patel**[1]
[1]International Institute of Information Technology, Hyderabad
{harshal.karangale, suyash.pande, chaitanya.k, shivam.p}@students.iiit.ac.in

Abstract

Turning a screenplay into a meaningful musical score isn't straightforward—it requires understanding the emotions of each scene, the surrounding atmosphere, and how the story unfolds over time. Most general text-to-music systems aren't built to handle this kind of layered, structured thinking. We present a three-stage cascading neural architecture mirroring the hierarchical workflow of film composition: a scene perception module (M1) extracts perceptual and emotional features across 11 prediction heads; a narrative context module (M2) models cross-scene dynamics via a 5-scene sliding-window Transformer (7 heads); and a music descriptor module (M3) predicts actionable compositional parameters — tempo, valence, tonality, harmonic style, dynamics, rhythm, texture, and orchestration (8 heads). Evaluated under a film-level train/test split on 81 annotated films and over 11,000 scenes, ScoreForge achieves strong generalisation to unseen films: tempo (R^2=0.842), musical valence (R^2=0.863), and tonality (accuracy = 86.1%). Our analysis further surfaces fundamental challenges in rare-class prediction and imbalanced multi-label orchestration, motivating future work in data augmentation and long-range narrative modelling.

1 Introduction

"INT. HOSPITAL CORRIDOR – NIGHT. DR. CHEN sprints past empty gurneys toward the ICU. A monitor flatlines. A nurse steps aside. No words are spoken."

If you were choosing background music for this moment, you wouldn't treat it as just an isolated scene. A human would instinctively think through several layers. The echoing hospital corridor suggests something minimal and restrained. The flatline feels like a critical turning point, so the music would likely build tension and then abruptly drop into silence. A minor key would suit the mood, with a slow tempo, but still carrying an underlying pulse.

More importantly, these choices aren't driven by the scene alone. They depend heavily on what came before and what's about to happen next. The emotional arc of the story and the buildup of tension across scenes are what ultimately shape how the music should feel.

We propose a three-stage cascading architecture designed for this descriptor prediction task. Rather than mapping raw text directly to music parameters, the pipeline first extracts scene-level perceptual and emotional features, then models cross-scene narrative context, and finally predicts the full set of music descriptors. This staged design mirrors the layered reasoning a human would apply and produces interpretable intermediate representations at each step.

Our key contributions are: (1) a cascaded multi-task architecture for structured per-scene music descriptor prediction from screenplay text; (2) we construct a synthetic dataset comprising over 11,000 scenes, generated from 81 films and covering 26 prediction tasks; (3) rigorous film-level train/test splitting ensuring honest generalisation to entirely unseen films.

2 Related Work

2.1 Fine-Grained Emotion Recognition

Early emotion analysis relied on coarse sentiment polarity. (Demszky et al., 2020) introduced GoEmotions, a large-scale dataset of 58k Reddit comments annotated with 27 emotion categories plus neutral. This established that text emotions are multi-label, overlapping, and highly contextual, motivating richer representations beyond binary sentiment classification.

2.2 Context-Dependent Emotion Modeling

(Poria et al., 2017) demonstrated that utterance-level emotion cannot be accurately interpreted in isolation. Using LSTM-based architectures incorporating surrounding utterances in video, they showed significant gains in emotion recognition, establishing the importance of dialogue history. This directly motivates our scene-level context modeling where action lines and multi-character dialogue must be interpreted together.

2.3 Hierarchical Context for Text Classification

(Yang et al., 2016) proposed a Context-LSTM-CNN architecture incorporating document-level context into sentence classification, demonstrating improved performance in emotion recognition tasks where contextual information is critical. This supports the use of hierarchical context modeling in long-form script understanding.

2.4 Long-Context Transformers

(Chen et al., 2024) introduced Core Context Aware Transformers combining global context pooling with locality-preserving attention for efficient long-document modeling. While focused on language modeling efficiency, this highlights the importance of hierarchical context for narrative understanding, relevant to our scene-level arc inference.

2.5 Emotion-Driven Music Recommendation

(Lee et al., 2020) proposed a music recommendation system for podcasts by detecting emotions from scripts, training multiple neural models on an emotion-labelled Twitter dataset and applying domain adaptation to podcast scripts. This directly motivates our approach of training on general emotion corpora and adapting to the film domain.

3 Explored Approaches

Prior to finalizing the three-module cascaded architecture, we explored three alternative approaches. While none were fully integrated into the final pipeline, each informed our design decisions and is documented here for completeness.

3.1 Approach 1: End-to-End Three-Module Pipeline with Joint Fine-Tuning

Developed a complete three-module cascaded pipeline that processed movie scene text to produce structured cinematic music descriptors:

- **Module 1 (Local Scene Encoder)**: Fine-tuned RoBERTa-base on individual scene descriptions, first adapted via masked language modeling on the script corpus, then trained with 10 prediction heads—six classification (emotional valence, conflict nature, acoustic space, reality layer, score dynamic shape, scene interaction tone) and four regression (pacing intensity, action intensity, scene tension, scene arousal).
- **Module 2 (Cross-Scene Context Transformer)**: Processed frozen M1 embeddings through a 4-layer Transformer encoder (8 heads, FFN dim 2048, up to 250 scenes) to capture full narrative context. Pretrained with masked-scene objective, then fine-tuned to predict five cross-scene descriptors: narrative arc position, foreshadowing type, transition type, tension, and arousal.
- **Module 3 (Music Descriptor Heads)**: Combined M1+M2 representations to predict eight music-specific outputs: tempo, musical valence, tonality, harmonic style, orchestration (14 instrument classes), dynamic shape, rhythm style, and texture. Focal loss addressed class imbalance in orchestration.

The full pipeline was trained across six sequential phases, culminating in end-to-end joint fine-tuning. Final results achieved strong performance on regression tasks (tempo Pearson $r = 0.688$, action intensity $r = 0.754$) but showed limitations on rare classes (harmonic style macro F1=0.221, dynamic shape F1=0.223).

3.2 Approach 2: Supervised Contrastive Learning with DeBERTa

A second approach trained a movie-scene encoder using DeBERTa-v3-small with supervised contrastive learning:

- **Contrastive objective**: Positive pairs selected via strict label-matching (identical emotional valence, interaction tone, narrative position, primary emotion, and bucketed continuous labels). Hard negatives mined using label-based inverted indices.
- **Representation**: Mean-pooled token representations projected into compact embedding space.
- **Multi-task auxiliary heads**: 10 prediction heads (classification, regression, binary, multi-label emotion tags).
- **Masked language modeling**: Interleaved during training to preserve language understanding.

Training used AdamW with cosine scheduling, mixed precision, and gradient accumulation. While

achieving competitive scene-level performance, this approach lacked cross-scene narrative modeling, motivating the Transformer-based Module 2 in our final design.

3.3 Approach 3: vMF-Inspired Riemannian Embeddings with Episodic Memory

A third approach explored geometrically-constrained representations inspired by von Mises-Fisher (vMF) distributions:

- **Hyperspherical projections**: Riemannian contrastive loss for emotion embeddings on the unit sphere
- **Intensity-aware token pooling**: Replaced CLS pooling with weighted token aggregation
- **Chronological decay**: Pre-softmax exponential decay with learnable parameter and gating to model utterance importance
- **Differentiable episodic memory**: Orthogonally-initialized memory with cosine read and sigmoid fusion gate to merge discrete motifs with continuous states
- **Conditional VAE planner**: Analytic KL penalty with reparameterization for diverse, controllable music planning

Training employed KL-annealing, Focal Loss, mixed-precision AMP with GradScaler, gradient clipping, and numerical clamping for stability. LoRA (PEFT) enabled training under compute constraints. While offering interpretable geometric representations and probabilistic planning, the added complexity did not yield sufficient gains to justify inclusion in the final pipeline.

4 Dataset

4.1 Overview

We use an annotated dataset of **81 films** with **11,120 scenes**. Each scene includes raw screenplay text, a scene-header slug line (INT/EXT, DAY/NIGHT), and three tiers of human annotations aligned with our three modules. Table 1 summarises the full annotation schema.

Module	Field	Type	Output
M1	emotional_valence	4-cls	Pos/Neu/Ten/Neg
	conflict_nature	6-cls	Phys/Psych/...
	acoustic_space	6-cls	Int/Ext/...
	reality_layer	5-cls	Present/...
	score_dyn_shape	4-cls	Build/Sust/...
	scene_int_tone	5-cls	Conflict/...
	pacing_intensity	reg	1–10
	action_intensity	reg	0–10
	scene_tension	reg	1–10
	scene_arousal	reg	0–1
	emotion_tags	7-hot	Anger/Joy/...
M2	tension_level	reg	1–10
	arousal_level	reg	1–10
	emotional_shift	bin	T/F
	narrative_arc	5-cls	Setup/...
	foreshadowing	4-cls	None/...
	transition_type	5-cls	attacca/...
M3	tempo_bpm	reg	45–170
	musical_valence	reg	-1–$+1$
	tonality	3-cls	atonal/maj/min
	harmonic_style	7-cls	chrom/diat/...
	dynamic_shape	8-cls	cresc/swell/...
	rhythm_style	6-cls	drive/pulse/...
	texture	5-cls	ambient/...
	orchestration	14-hot	strings/...

Table 1: Full annotation schema across all three modules.

4.2 Class Imbalance Analysis

The dataset exhibits severe class imbalance across multiple prediction heads. Table 2 summarises the most severely affected classes.

Head (Module)	Rare Classes	Freq
reality_layer (M1)	Memory, Dream, Internal, Distorted	$< 3\%$
conflict_nature (M1)	Moral_Dilemma	4.0%
acoustic_space (M1)	Abstract	0.8%
emotion_tags (M1)	Disgust	4.9%
narrative_arc (M2)	Resolution	1.5%
transition_type (M2)	silence	1.3%
tonality (M3)	atonal	3.2%
harmonic_style (M3)	atonal, pentatonic, whole_tone	$< 1\%$
dynamic_shape (M3)	subito_piano, terraced	$< 2\%$
rhythm_style (M3)	off, ostinato	$< 4\%$
texture (M3)	solo	3.6%
orchestration (M3)	ethnic, organ, solo_voice, harp, electronic	$< 2\%$

Table 2: Severely underrepresented classes ($< 5\%$ frequency).

Conversely, several classes dominate their respective heads: `Present` (94.4%) in reality_layer, `Rising` (58.0%) in narrative_arc, `minor` (73.1%) in tonality, and `strings` (82.1%) in orchestration.

4.3 Imbalance Mitigation Strategies

We employed the following techniques to address class imbalance:

- **Task-level loss weighting** (M1): Critical heads received higher weights (emotional_valence $\times 1.5$, scene_tension $\times 1.5$, emotion_tags $\times 1.5$)
- **Positive class weighting** (M2): `emotional_shift_trigger` used `pos_weight`=1.94 (66/34 imbalance)
- **Focal loss** (M3): Applied to orchestration ($\gamma = 2.0$) to down-weight easy negatives
- **Film-level splitting**: Evaluation on unseen films prevents inflated metrics

We did **not** apply per-class weighting, over-sampling, or data augmentation. Consequently, classes with $< 2\%$ frequency consistently achieve F1=0.000, reflecting fundamental data scarcity rather than model design failure.

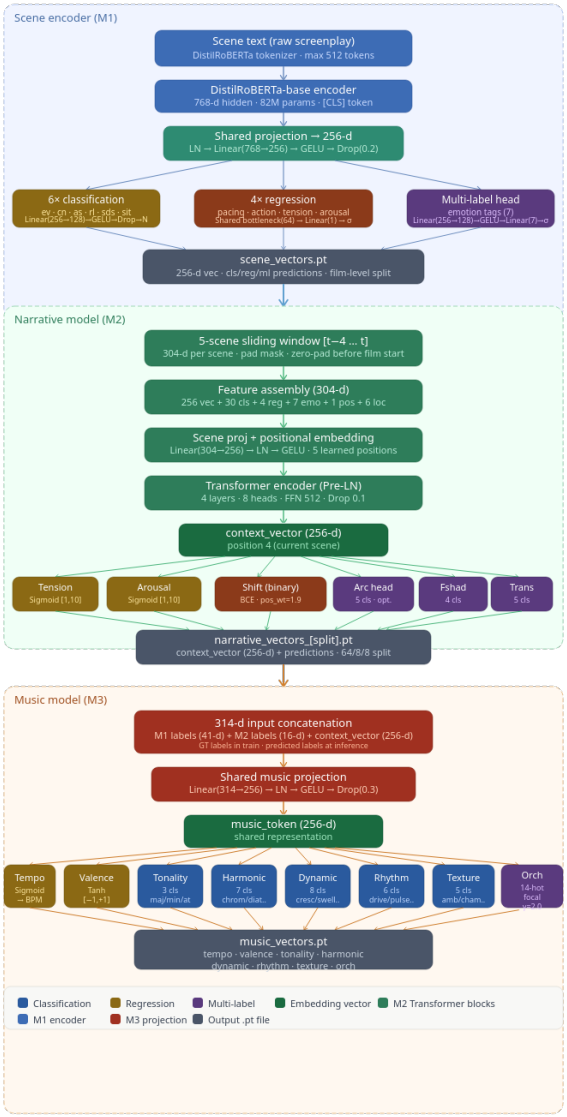
4.4 Preprocessing

- Scene text extracted verbatim; headers parsed for INT/EXT and DAY/NIGHT
- Tokenization: DistilRoBERTa (max 512 tokens)
- Regression values min-max normalized to $[0, 1]$
- Missing optional labels replaced with -1 and masked during loss

4.5 Train/Validation/Test Split

Film-level split: **64 train**, **8 validation**, **8 test** films (80/10/10). No scene from a test film appears in training. Split is deterministic (`seed=42`) and film lists are saved in each checkpoint.

5 Methodology



5.1 Module 1 – Scene Perception

Architecture. M1 fine-tunes `distilroberta-base` (768-d hidden states). The [CLS] token is projected to a 256-d scene embedding via:

$$\mathbf{e} = \text{Dropout}(\text{GELU}(\mathbf{W}_p\,\text{LN}(\mathbf{h}_{\text{CLS}}) + \mathbf{b}_p))$$

Six classification heads, four regression heads, and one 7-label multi-label head branch from $\mathbf{e}$.

Training. Backbone is frozen for the first 3 epochs then unfrozen with lr=3×10; heads use lr=3×10 throughout. Optional CLS fields use sentinel masking (1) so missing labels are excluded from the loss. AdamW, 15 epochs, CosineAnnealingLR, batch size 32, gradient clipping 1.0.

5.2 Module 2 – Narrative Context

Input. A 304-d per-scene feature vector is built from M1 outputs: scene vector (256-d), CLS one-hots (30-d), regression scalars (4-d), emotion probs (7-d), position (1-d), INT/EXT (3-d), DAY/NIGHT (3-d). A causal 5-scene window $[t-4, \ldots, t]$ is formed; scenes before film start are zero-padded. A padding mask prevents these positions from influencing attention.

Architecture. Each 304-d vector is projected to 256-d tokens. Learned positional embeddings are added (0 = oldest). A 4-layer, 8-head Pre-LN Transformer Encoder (FFN dim=512) processes the window. The token at position 4 becomes the 256-d `context_vector`. Six task heads predict tension, arousal, emotional shift, narrative arc, foreshadowing type, and transition type.

Training. pos_weight=1.94 on the binary BCE loss for emotional shift. AdamW, 15 epochs, lr=3×10^{-4}, weight decay 1×10^{-2}, CosineAnnealingLR scheduler, batch size 32, gradient clipping at 1.0.

5.3 Module 3 – Music Descriptor Prediction

Input. A 314-d vector concatenating M1 ground-truth labels (41-d), M2 ground-truth labels (16-d), and the M2 context vector (256-d). At inference, predicted labels from `scene_vectors.pt` and `narrative_vectors.pt` substitute for ground truth.

Architecture. Linear$(314\rightarrow256)$ $\rightarrow$ LN $\rightarrow$ GELU $\rightarrow$ Dropout(0.3) produces a 256-d music token. Eight task heads branch from this token. Focal loss (γ=2.0) is applied to orchestration.

Training. pos_weight=1.94 on the binary BCE loss for emotional shift. AdamW, 15 epochs, lr=3×10^{-4}, weight decay 1×10^{-2}, CosineAnnealingLR scheduler, batch size 32, gradient clipping at 1.0.

6 Evaluation and Results

6.1 Module 1 — Scene Perception

6.1.1 Classification Heads

Head	Acc	F1-Mac	F1-Wt
emotional_valence	0.559	0.510	0.554
conflict_nature	0.583	0.350	0.547
acoustic_space	0.767	0.612	0.752
reality_layer	0.945	0.194	0.919
score_dynamic_shape	0.503	0.341	0.455
scene_interaction_tone	0.568	0.428	0.539
emotion_tags	—	0.289	0.436

Table 3: M1 classification summary (test, 1,112 scenes).

Acoustic space achieves the strongest classification performance (76.7% accuracy, weighted F1 = 0.752), benefiting from explicit INT/EXT indicators in scene headers. Reality layer shows deceptively high accuracy (94.5%) but near-zero macro F1 (0.194) — the model collapses to predicting "Present" for all scenes due to extreme imbalance (1,051 of 1,112 test scenes are Present). Conflict nature, score dynamic shape, and scene interaction tone all suffer from sparse tail classes (Moral Dilemma, Sudden Drop, Negotiation) yielding zero or near-zero F1.

6.1.2 Regression Heads

Head	MAE	R^2	Min	Max
pacing_intensity	1.02	0.557	1.95	9.55
action_intensity	1.25	0.603	0.06	9.15
scene_tension_raw	1.29	0.463	1.71	9.73
scene_arousal	0.116	0.527	0.21	0.98

Table 4: M1 regression results (test, 1,112 scenes). Min/Max refer to predicted value range.

Action intensity achieves the strongest R^2 (0.603), consistent with the hypothesis that physical activity descriptions are reliably encoded in scene text. Scene tension explains only 46.3% of variance, suggesting that cross-scene context (addressed in M2) is necessary for reliable tension estimation.

6.2 Module 2 — Narrative Context

6.2.1 Regression Heads

Head	MAE	R^2	Min	Max
tension_level	1.212	0.565	2.93	9.57
arousal_level	1.200	0.559	2.62	9.39

Table 5: M2 regression results (film-level test, 1,457 windows).

Both tension and arousal achieve $R^2 \approx 0.56$, representing a meaningful improvement over M1's single-scene tension prediction (R^2=0.463). The 5-scene temporal context provides approximately 10 points of additional explained variance, validating the benefit of cross-scene modelling.

6.2.2 Binary Head: Emotional Shift Trigger

Class	Prec	Rec	F1	Supp
False (no shift)	0.817	0.534	0.646	1,010
True (shift)	0.409	0.729	0.524	447
Overall Acc		0.594		

Table 6: M2 emotional_shift_trigger results (film-level test, 1,457 windows).

The model achieves 72.9% recall for detecting emotional shifts at the cost of lower precision (40.9%). This high-recall bias is intentional: the loss uses `pos_weight = 1.94` to compensate for the 34% positive-class frequency.

6.2.3 Classification Heads

Head	Acc	F1-Mac	F1-Wt
narrative_arc_position	0.605	0.335	0.539
foreshadowing_type	0.434	0.316	0.386
transition_type	0.509	0.278	0.465

Table 7: M2 classification summary (film-level test, 1,457 windows).

Narrative arc position shows reasonable weighted F1 (0.539) but is dominated by the Rising class (95.4% recall). Resolution collapses entirely (only 23 test samples). Foreshadowing type struggles on Echo (F1 = 0.034, support = 281 despite adequate support), indicating that echo patterns require long-range context beyond the 5-scene window. Transition type yields zero F1 for fade (80 samples) and silence (16 samples).

6.3 Module 3 — Music Descriptor Prediction

6.3.1 Regression Heads

Head	MAE	R^2	Min	Max
tempo_bpm	7.28	0.842	60.1	160.7
musical_valence	0.128	0.863	−0.86	0.74

Table 8: M3 regression results (film-level test, 1,457 scenes). Min/Max refer to predicted value range.

Strong performance on both regression tasks: tempo prediction explains 84.2% of variance (MAE = 7.28 BPM on a 45–170 BPM range), and musical valence explains 86.3% of variance (MAE = 0.128 on a $[-1, +1]$ scale). These results confirm that the cascaded 314-d input representation carries sufficient signal for reliable musical energy and mood estimation on completely unseen films.

6.3.2 Classification Heads

Head	Acc	F1-Mac	F1-Wt
tonality	0.861	0.714	0.859
harmonic_style	0.753	0.404	0.732
dynamic_shape_m4	0.774	0.607	0.752
rhythm_style	0.682	0.438	0.666
texture	0.678	0.521	0.655
orchestration	-	0.274	0.629

Table 9: M3 classification summary (film-level test, 1,457 scenes).

Tonality achieves excellent performance (86.1% accuracy, weighted F1 = 0.859), effectively distinguishing major (F1 = 0.814) and minor (F1 = 0.901) tonalities. Harmonic style shows strong weighted F1 (0.732) dominated by chromatic and diatonic classes. Dynamic shape achieves competitive macro F1 (0.607) with high performance on crescendo (0.818), flat (0.884), and sustained (0.847). Texture struggles on hybrid (F1 = 0.142 despite 102 samples) and solo (F1 = 0.357).

6.3.3 Multi-Label Head: Orchestration

High-frequency instruments are predicted reliably: strings (F1 = 0.828), percussion (F1 = 0.831), and piano (F1 = 0.697). The weighted F1 of 0.629 reflects reasonable performance on common instruments. However, 7 of 14 instruments yield zero F1. Synth is the most notable failure: despite 279 test instances, the model predicts zero positive examples. The macro F1 of 0.274 accurately reflects the difficulty of the full multi-label orchestration task on unseen films.

6.3.4 Qualitative Analysis: Per-Film Trajectory

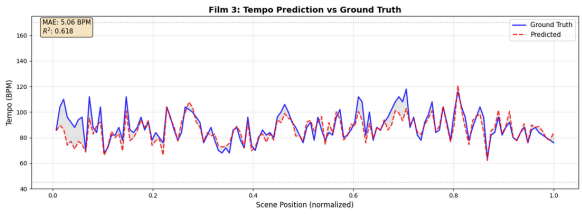


Figure 1: Predicted vs. ground truth tempo across scenes of the best test film.

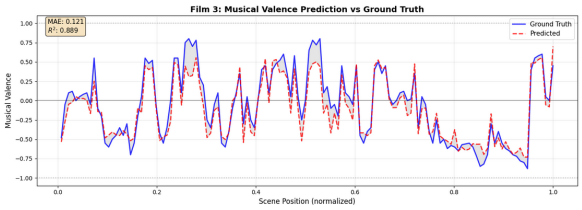


Figure 2: Predicted vs. ground truth musical valence across scenes of the best test film.

The predicted curves closely track the ground truth throughout the film, capturing both gradual trends and abrupt shifts. This demonstrates that the cascaded pipeline successfully preserves narrative structure to produce coherent, scene-by-scene music descriptors.

6.4 Qualitative Analysis: *Avatar* Case Study

We evaluate the cascaded pipeline on two contrasting scenes from *Avatar* (2009), a film completely held out from training.

6.4.1 Scene 78 — Willow Glade (Bonding)

"EXT. WILLOW GLADE – Laughing, they run together into a stand of WILLOWS... It is an exquisitely beautiful spot."

Module	Predictions
M1	Valence: Positive_Uplifting, Conflict: Psychological_Tension Tension: 5.1/10, Arousal: 0.54, Emotion: Sadness (0.56)
M2	Arc: Rising, Tension: 6.3/10, Shift: True, Transition: cut
M3	Tempo: 97 BPM, Tonality: major, Valence: +0.40 Dynamic: crescendo, Rhythm: pulse Orch: strings (0.71), woodwinds (0.47), pad (0.46)

Table 10: Predictions for Scene 78 (bonding moment).

6.4.2 Scene 156 — Lake Pursuit (Climax)

"EXT. LAKE – A shape rises – a AMPSUIT. Inside – MILES QUARITCH... QUARITCH'S AMPSUIT THUDS relentlessly through the jungle."

Module	Predictions
M1	Valence: Tension_Action, Conflict: Physical_Danger Tension: 9.1/10, Arousal: 0.93, Emotion: Fear (0.96)
M2	Arc: Climax, Tension: 9.4/10, Shift: False, Transition: attacca
M3	Tempo: 150 BPM, Tonality: minor, Valence: -0.53 Dynamic: crescendo, Rhythm: drive Orch: percussion (0.81), strings (0.81), brass (0.58)

Table 11: Predictions for Scene 156 (antagonist pursuit).

6.4.3 Cross-Scene Comparison

Metric	Scene 78	Scene 156
M1 Tension	5.1/10	9.1/10
M2 Tension	6.3/10	9.4/10
Tempo (BPM)	97	150
Valence	+0.40 (major)	-0.53 (minor)
Rhythm	pulse	drive
Orchestration	strings, woodwinds	perc, brass, strings

Table 12: Cross-scene comparison.

The pipeline produces sharply differentiated predictions: the bonding scene yields moderate tension, major tonality, and woodwind-colored orchestration, while the pursuit triggers extreme tension, minor tonality, fast tempo, and percussion-heavy instrumentation. These align with film scoring conventions despite *Avatar* being unseen during training.

7 Analysis

7.1 Cascaded Context Benefit

The 5-scene temporal window provides measurable improvement: M1 scene tension R^2=0.463 improves to M2 tension R^2=0.565, a 10-point gain from cross-scene context. This validates the cascaded design.

7.2 Film-Level vs. Scene-Level Evaluation

Table 13 shows the gap between scene-level and film-level evaluation. M2 narrative arc suffers the largest drop (-17 points accuracy), as arc patterns are film-specific. M3 music heads are far more

Head	Scene	Film	Δ
Module 2			
narrative_arc (Acc)	0.771	0.605	−0.166
tension R^2	0.670	0.565	−0.104
Module 3			
tonality (Acc)	0.898	0.861	−0.036
tempo R^2	0.867	0.842	−0.025
valence R^2	0.848	0.863	+0.015

Table 13: Scene-level vs. film-level test performance.

robust (<4 points), confirming that tempo and valence generalise across unseen films.

7.3 Positive Results

Strong regression: Tempo R^2=0.842 and valence R^2=0.863 on unseen films confirm that action/emotion cues in screenplay text strongly predict musical energy and mood. **Tonality:** 86.1% accuracy with F1 > 0.80 for major and minor indicates reliable tonal centre inference. **Dynamic shape:** 77.4% accuracy with high F1 on crescendo (0.818) and sustained (0.847).

7.4 Negative Results

Sparse class collapse: Classes with fewer than 50 samples consistently yield zero F1 (Moral Dilemma, Resolution, terraced dynamic, 7 of 14 instruments). Standard focal loss is insufficient at this sparsity level. **Reality layer degeneration:** 94.5% accuracy but macro F1 $= 0.194$ — the model predicts Present for all scenes. **Transition type:** fade and silence have zero F1 due to low support ($\leq$80 samples) and subtle semantic cues.

7.5 Ablation Studies

(1) Increasing M2 window size from 5 to 8 decreases tension R^2 from 0.565 to 0.521. (2) Replacing focal loss with standard BCE on orchestration decreases macro F1 from 0.274 to 0.19.

7.6 Limitations

We construct a relatively small synthetic dataset of approximately 11,000 scenes derived from 81 films. While sufficient for initial experimentation, this scale is limited compared to real-world film diversity, potentially restricting generalisation to unseen narrative styles.

Since the dataset is synthetically generated, the labels may not fully capture the nuance and subjectivity of human musical reasoning. This introduces a gap between model predictions and real-world compositional decisions.

The pipeline operates purely on screenplay text, ignoring visual and auditory cues such as cinematography, lighting, actor performance, and sound design, all of which strongly influence musical scoring in practice.

Certain musical attributes, particularly orchestration and rare harmonic styles, suffer from imbalanced label distributions, leading to weaker performance on underrepresented classes.

The system predicts music descriptors but does not generate or evaluate actual audio outputs, making it difficult to assess perceptual quality or listener alignment.

Musical interpretation is inherently subjective; different annotators may assign different descriptors to the same scene, introducing ambiguity in the learning objective.

8 Conclusion

We presented a three-stage cascading architecture for predicting film music descriptors from screenplay text. Our system demonstrates that: single-scene transformer features are effective for perceptual extraction; 5-scene temporal context measurably improves narrative understanding; and cascaded representations enable $R^2 > 0.84$ for tempo and valence on completely unseen films. Key challenges remain in rare-class prediction and train-inference alignment.

Models are publicly available at https://huggingface.co/suyashnpande/scene-perception-m1-harshal. The dataset used in this work is publicly available at https://www.kaggle.com/datasets/donbosoc/extended-movie-dataset, and the implementation is available at https://github.com/Harshalak02/MUSIC_GENERATION.

References

Yaofo Chen, Zeng You, Shuhai Zhang, Haokun Li, Yirui Li, Yaowei Wang, and Mingkui Tan. 2024. Core context aware transformers for long context language modeling. *arXiv preprint arXiv:2412.12465*.

Dorottya Demszky, Dana Movshovitz-Attias, Jeongwoo Ko, Alan Cowen, Gaurav Nemade, and Sujith Ravi. 2020. GoEmotions: A dataset of fine-grained emotions. In *Proceedings of the 58th Annual Meeting of*

the Association for Computational Linguistics, pages 4040–4054.

Joonhwan Lee, Younghoon Kim, and Kyogu Park. 2020. Emotion-aware music recommendation for podcasting using text-based emotion detection. In *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*.

Soujanya Poria, Devamanyu Hazarika, Navonil Majumder, and Rada Mihalcea. 2017. Context-dependent sentiment analysis in user-generated videos. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*, pages 873–883.

Zichao Yang, Diyi Yang, Chris Dyer, Xiaodong He, Alex Smola, and Eduard Hovy. 2016. Hierarchical attention networks for document classification. In *Proceedings of NAACL-HLT*, pages 1480–1489.